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Teaching Human-Computer Interaction with the use of Generative Artificial Intelligence Design Tools and Prototypes

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Abstract

Generative Artificial Intelligence (GenAI) has had a significant impact on education. Students have adapted and developed novel strategies and behaviors for learning and completing assignments with the assistance of AI. In a Human-Computer Interaction (HCI) class, we conducted an analysis of student behavior across two distinct academic years (2025 and 2026). Participants were granted unrestricted access to AI tools while working on their student projects. The findings revealed that students exhibited a progressive evolution in their methodologies, becoming selective about the tools they employ, the tasks they address with AI, and the methods they utilize to validate outputs and mitigate the occurrence of hallucinations. Furthermore, we assigned volunteers the task of utilizing AI-generated custom design tools instead of existing software to create their deliverables. During the assessment of the System Usability Scale (SUS), the final metrics and ratings (with a mean score of 85.20, corresponding to a Grade A/Excellent rating) demonstrated that the AI-generated tools outperformed their existing counterparts. These results position our research study as a pivotal point in the learning of HCI with AI, and higher education in general.

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Keywords: teaching; students; human-computer interaction; artificial intelligence; design tools; software; evaluation

1. Introduction

Artificial Intelligence (AI) in general and Generative AI (GenAI) in particular are evolving rapidly, moving beyond the perception of a simple automation script and transitioning towards a consistent source of collaboration and assistance [17]. The same applies to education, where students break away from familiar syntax or repetitive tasks and switch to AI to perform faster and manage complex tasks with minimal effort [18]. While this opportunity is a preferred choice for students, it does come with challenges. For example, students can use familiar AI tools to produce uninspired solutions or designs, resulting in a loss of originality [16]. Given the provided context, our research study aims to address the following inquiries: (1) How do students use AI tools when it comes specifically to User Interface

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(UI) design projects? (2) How can AI encourage learning motivation and creative exploration when assimilating new concepts from knowledge areas such as HCI? (3) Do quick AI-generated tools perform better than existing software used in the learning process(es)?

Paper structure: we first present our current findings in relation to the State of the Art (using AI in education, especially within HCI, and the use of AI tools in education – Section 2). Then, in Section 3 our proposal is presented, followed by the complete methodology we applied (Section 4) and the results we gained (Section 5). We also discuss the limitations and future work in Section 6, ending with conclusions (Section 7) and acknowledgments (Section 8).

2. State of Art

Current efforts to integrate GenAI into education [11, 12], particularly in the HCI field [13, 14], encompass various approaches and have yielded promising outcomes. With *AI as a Persona* (used for brainstorming and developing project ideas/scenarios) and *AI as a Mirror* (used for reflecting on student understanding and identifying knowledge gaps), students expressed increased engagement during class [1]. Other studies showcase the role of AI as a possible replacement for human researchers and participants with the right fine-tuned "prompt engineering" [2, 16].

Alternative investigations highlight the role of AI in generating highly correlated synthetic user research data that can serve as a proxy for human participants in HCI studies [4]. As impressive as AI results could be, human validation is necessary, especially when dealing with hallucinations (AI should not be treated as a "black box" without designer intervention [3]). Now, it is of the utmost importance, as design deliverables created with AI (generating user personas, UX copywriting, storytelling or HTML/CSS code for Web UI prototyping) can become part of the design process [5, 7]. Furthermore, GenAI is increasingly being adopted by UX professionals, and the challenges it presents to this new audience include maintainability, consistency across screens, and adherence to specific design systems [6]. With the transition into AI, new paradigms and frameworks are emerging that guide practitioners toward a responsible and ethical Human-Centered AI (HCAI), pushing industry themes from UX 1.0 (usability) and UX 2.0 (Internet/mobile) toward a new UX 3.0 (HCAI) [8].

3. Proposal

We structured our proposal into two stages to test whether AI-generated design tools can aid the UI design process. First, we collected insightful data on designing UI projects by using GenAI without restricting the process. Secondly, utilizing the collected feedback, we developed our own set of AI-generated design tools and conducted tests to evaluate their utility and usability. The AI tools developed were specifically designed to assist in the creation of design deliverables such as storyboards, user flows, sitemaps, digital sketches, and other related content. Moreover, for this iteration of our research study, the validation audience consists of last-year university students with basic design knowledge and experience. We also gathered data on how students use and apply AI in their university activities.

Our motivation for employing this two-layered structure in our study is to establish a foundational understanding, gain insights from it, and subsequently introduce new levels and proposals upon it.

Initially, we observed how students utilized AI in their projects and subsequently proposed novel tools (developed with AI) for their use. In the following sections, we will detail the methodology applied and the results achieved.

4. Methodology

Our research methodology spans two academic years (2025 and 2026). Figure 1 highlights a quick overview of the entire process.

In the first year (2025), HCI students worked on their design projects, building the required deliverables (e.g., persona profiles, storyboards, user flows, sitemaps) for final evaluation. We permitted the students to utilize AI freely during the development of their projects, subject to the following constraints: (1) Students were required to adhere to ethical guidelines when employing AI; (2) Students were responsible for verifying the output generated to prevent the occurrence of generative hallucinations [15].

After they finished their projects, we administered an online questionnaire to volunteers to gain further insights into their utilization of AI in their work and to solicit their perspectives on the application of (Generative) AI tools within academic institutions.

A year later, in 2026, the experiment was repeated with several major changes. We permitted the students to continue using AI for their project and to use our custom AI-generated design tools we provided. They would use these tools to refine their HCI project deliverables such as persona profiles, storyboards, user flows, sitemaps. After the students tested the tools, we invited them to volunteer as evaluators. These evaluators would utilize custom criteria and metrics to assess the tools' effectiveness. Additionally, to further validate the application of our tools, we solicited students to assess existing design tools available in the market. This evaluation would serve as a benchmark for comparing approaches, specifically between AI-generated and existing tools. Lastly, we conducted a repeat questionnaire on the utilization of AI in academia to compare changes and advancements since the previous year.

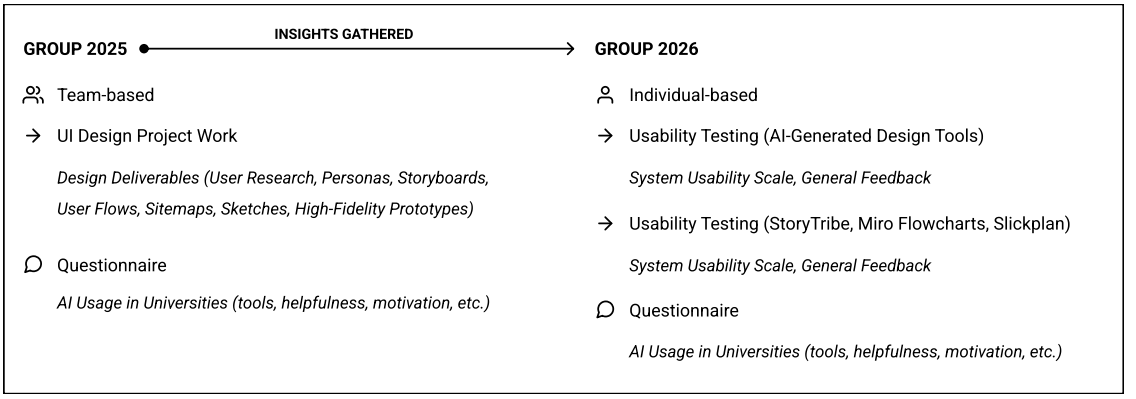


Fig. 1: Summary of our research validation process split into two student groups across two academic years (2025 and 2026)

4.1. Group 2025: Student Design Deliverables with AI

For the first group (2025), we recruited 75 last-year students enrolled in HCI classes (39 female and 36 male). Out of all participants, 20 were employed in junior-level positions (26.67%).

As part of the HCI class, the students had to create UI designs for multiple environments/devices – Web/desktop, mobile, smartwatch, Mixed Reality (MR), etc. – varying in topics: from dog training platforms and airsoft MR assistance, to first responder augmented devices, wellness applications, LEGO virtual guide or home automation ecosystem (an example is depicted in Figure 2). As deliverables, the assets varied, but they remained interconnected: students had first to conduct user research using different approaches, define their project's user persona, and then compile solutions using ideation techniques. Afterwards, they document the chosen solution using storyboards, user flows, and sitemaps. After creating low-fidelity sketches and setting up a style guide, they would build highly interactive working prototypes. For their final assignment, students were required to present their solution, accompanied by a video demonstration and comprehensive documentation.

4.2. Group 2025: Questionnaire

Since focusing on how students use AI during their university projects, we decided to evaluate their behavior and motivation in using such tools. Below are the main questions we asked the participants during our validation process:

- *How do you decide whether to use AI or not for a task?*
- *How often did you use AI tools for your HCI project?*
- *How helpful was AI in completing your project?*
- *What AI tools did you use for your project assignment(s) – e.g. ChatGPT, Claude, Perplexity?*
- *For which tasks did you use AI the most?*

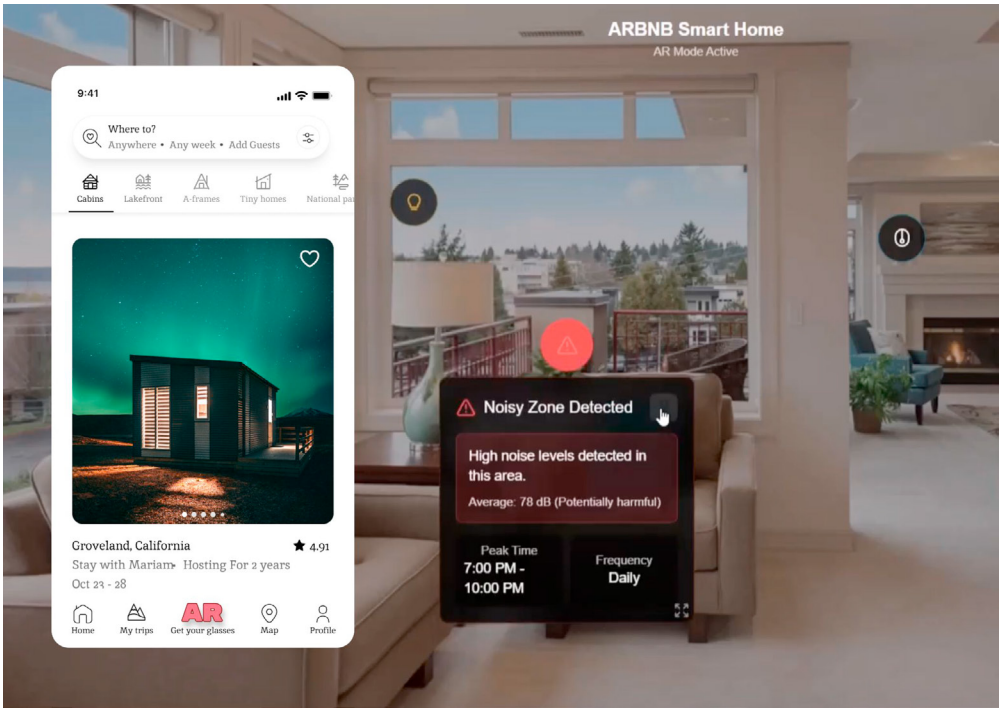


Fig. 2: Example of a team's project prototype in the home automation theme and topic

- *How did you use AI for the options selected above?*
- *What was the biggest advantage of using AI in your HCI project? And the biggest downside?*
- *Should schools/universities provide Generative AI training for students? Please explain.*

4.3. Group 2026: New AI Tools

For the second group (2026), we recruited 66 last-year students from HCI classes (13 female, 53 male), of whom 19 (28.79%) were employed for the first time in their careers. Table 1 summarizes the audience for both groups.

Table 1: HCI Audience

Group	Participants	Employed	Male	Female
2025	75	20 (26.67%)	36	39
2026	66	19 (28.79%)	53	13

As previously stated, we built dedicated applications that help students create directly in the Web browser design deliverables for their HCI projects. Following last year's structure, we focused on the main project requirements. We used v0.app[1] to speed up creation and ensure we have high-fidelity interactive prototypes that mimic real-life design use cases to a great degree. From our experience testing a diverse range of AI tools, v0 proved the most effective and accurate prompt request, yielding high-quality results.

We build each prototype using step-by-step instructions: first starting with a detailed prompt containing the major insights from the first group, then gradually iterating until we had a complete working flow inside the prototype without any bugs or application crashes. For performance and simplicity reasons we kept the prototypes as client-facing applications. The following list details each GenAI tool created:

[1] v0 by Vercel – https://v0.app/ (Last accessed: May 15th, 2026)

- **HCI Research**: create and manage user research insights (literature review, competitor analysis, observational research, and more).
- **HCI Idea**: create and manage the ideation process [20]: from clusters and statements to How-Might-We(s) (HMW), brainstorming, and voting solutions.
- **HCI Persona**: a micro application to fill in the user persona profile.
- **HCI Story**: create and manage user storyboards.
- **HCI Flow**: create and manage user-flows (see Figure 3, right side).
- **HCI Map**: create and manage application sitemaps (see Figure 3, left side).

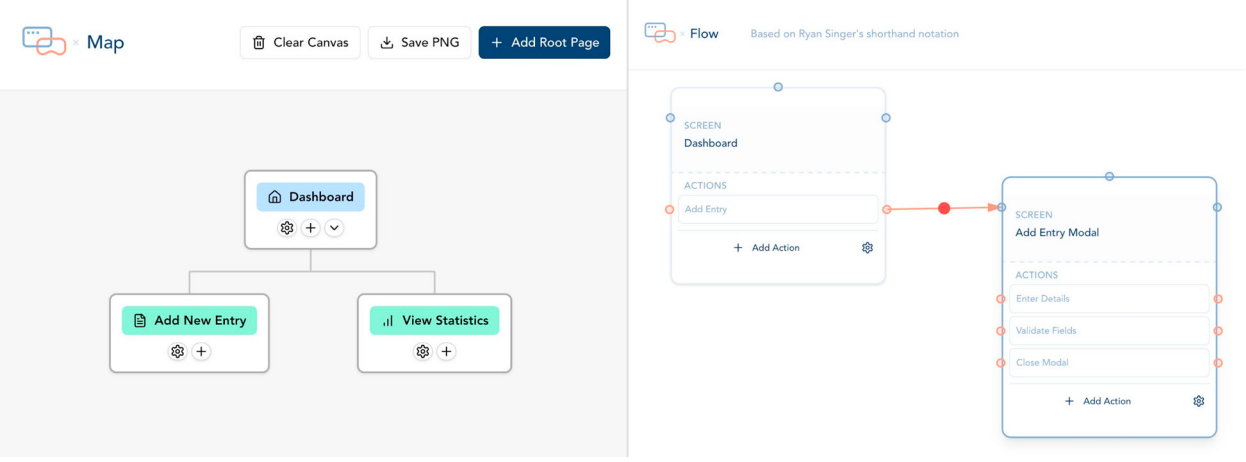


Fig. 3: Example of AI Generated User Sitemap (Left) and User Flow (Right) Design Tools using v0

Additional tools were implemented to perform UI sketches, manage style guides and animate image assets inside a simulated 3D world.

4.4. Group 2026: Testing/Evaluating the AI Tools

To validate our proposed AI-generated design tools, we invited participants to test three prototypes of their choice from the available stack provided by us (revisit list from Subsection 4.3), create deliverables for their projects using them, and evaluate them by answering specific follow-up questions. Each volunteer had 90 minutes at disposal to complete the tasks. The main metric we measure is the System Usability Scale (SUS) [9], as it provides a strong basis for benchmarking the usefulness and ease of interaction of our prototypes. By calculating the SUS values, we can compare it with industry defaults and with other design tools students might use.

4.5. Group 2026: Testing/Evaluating the Existing Design Tools

One key aspect of our validation process is testing existing design tools that produce deliverables similar to our AI-generated prototypes. We selected compatible applications based on our experience and tool popularity, and we decided on StoryTribe[2] (for creating storyboards), Miro Flowcharts[3] (for crafting user flows), and Slickplan[4] (for organizing sitemaps). Participants had the same amount of time and the same type of tasks to perform. We chose this strategy to ensure a 1:1 comparison between the two sets of applications (from features and capabilities to SUS and other types of feedback).

[2] StoryTribe: Free Online Storyboard Maker for Professionals – https://storytribe.com/ (Last accessed: May 15th, 2026)
[3] Flowcharts Online by Miro – https://miro.com/flowchart/online/ (Last accessed: May 15th, 2026)
[4] Slickplan: Website Planning Tools – https://slickplan.com/ (Last accessed: May 15th, 2026)

4.6. Group 2026: Questionnaire

Besides testing our design tools and the existing tools, we also invited the 2026 group to share their thoughts and experience regarding using AI in education. We decided to reuse the same type of questions we used with Group 2025 (Section 4.2) to have a clearer picture and comparison between groups and years.

5. Results

5.1. Group 2025: Questionnaire

The first set of quantitative results we calculated was the level of helpfulness AI provides and the impact AI has on the motivation to learn (using 1 to 5 ratings). Based on the students' ratings, we calculated averages of 3.44 for Helpfulness and **3.24** for Motivation Impact. As of early 2025, students expressed uncertainty and indecision about AI adoption due to a lack of clear standards and the constant shifts in technology trends. This insight is corroborated by other feedback they provided.

Students mainly use AI for repetitive tasks, writing documentation, and formatting tables and other data. AI is useful when students begin a task from scratch and need a starting point to spark their inspiration and motivation. On the other hand, those who do not use AI criticize the lack of originality in AI-generated content and the concerns about privacy and data usage.

An overwhelming majority of students chose "Occasionally" as to how frequently they use AI. They do not rely on AI as a constant solution, but rather use it as a specialized tool for specific friction points. These findings are consistent with other insights indicated by similar studies such as [19].

Over 54% of Group 2025 rated AI as 3 or 4 for its helpfulness. This percentage suggests that, for the majority, AI is a reliable "assistant" that helps them move forward, but the student remains the primary driver.

The top three AI assistants are OpenAI ChatGPT, Anthropic Claude, and Google Gemini. For UI/UX-specific tasks, Vercel v0 is the preferred choice, while DeepSeek and GitHub Copilot are better for programming. For research, students choose Perplexity. Midjourney or similar tools are considered good alternatives for visual assets.

The main HCI project tasks that users relied on AI to complete were user persona creation and project documentation. For creative assets such as prototypes, style guides, or storyboards, students are likely using AI to generate the logic and text (placeholders, button labels, user flows). They prefer doing the manual design themselves to maintain the "personal touch" they mentioned in earlier answers. Interestingly, students still recognize that *user research and testing require real human interaction*.

Students start with a basic prompt, filter the results, and refine the details afterward. For style guides, they generated color palettes and fonts while checking the contrast themselves. For research, participants identified competitors and summarized literature reviews using AI, but personally validated the existence of the resources. For more complex tasks, such as working on AR/VR features, they used AI to brainstorm ideas, but left the final decision to the human team. For prototyping, students used v0 to generate basic UI layouts, while keeping the personalization to do themselves in Figma, a popular collaborative UI design tool.

As main advantages, the volunteers reported a significant increase in speed when building prototypes and exploring different solutions before committing to a final approach. On the other hand, the main disadvantages/complaints include the *frequency of hallucinations* and the need to constantly *fact-check and correct the output*, or provide additional detailed context, which impacts the overall effort.

As for how the university can contribute with AI adoption in education, surprisingly, students did not promote the need for a dedicated course in their curriculum on how to use ChatGPT or other tools; rather, they need more on how to spot hallucinations and "confidently wrong" answers, understanding plagiarism, data privacy, and environmental impact, and knowing when not to use AI (e.g., for legal/medical advice or core creative tasks). Moreover, almost all students agree that universities should integrate AI into their education programs. They view it as professional preparation, fearing they'll be left behind if the university ignores it since companies already use it.

Nearly a third of the audience feels more motivated by AI, but 52% believe it has no significant impact on their motivation to learn. This observation, supported by the mean motivation score, , provides valuable insights for educators: as of 2025, it suggests that, *for most students, AI is a "utility" rather than a "transformative experience"*.

5.2. Group 2026: Testing/Evaluating AI-generated Design Tools

We extracted the remaining quantitative data from students' evaluations of their AI tools and calculated SUS values for each (see Table 2). All of our proposed AI-generated tools achieved a SUS value of 80 or higher (Grade A/"Excellent") with an average of 86.44.

Table 2: SUS values for AI-generated design tools

Tool	Research	Idea	Persona	Story	Flow	Map
Mean SUS	89.11	80.91	91.03	84.24	84.93	86.41

The top three highest-rated tools are HCI Persona (**91.03**), HCI Research (**89.11**), and HCI Map (**86.41**). Students appreciated the clear, quick, and simple approach to finalizing their task when creating a persona template, managing all their user research data, and building complex sitemaps.

At the opposite end, tools such as HCI Flow (**84.93**), HCI Story (**84.24**), and HCI Idea (**80.91**) require an additional learning curve and time to get used to the concepts presented. In the following paragraphs, we present the main feedback and observations the students had for our AI-generated tools:

- Users perceived the **HCI Flow** tool as "easy to use" and "visually clear". They understood the drag-and-drop mechanics immediately, with no steep learning curve. In addition to enhancing user experience, users have indicated a preference for an infinite canvas, the capability to automatically align or snap to a grid to maintain flow organization, and the option to control the z-index or configure dark mode.
- Users appreciated that the **HCI Idea** tool used linear progression (from Insights to Voting), breaking the complex ideation process into simple steps. The clean, simple UI helped participants focus without being visually overwhelmed. The only major downsides were the need for more context explanations to newer terms, such as "How Might We" questions or "Clusters".
- Volunteers considered the **HCI Map** tool the most "polished" prototype. The clean, minimalist interface conveys a sense of "utility" and attention to detail, such as the disclaimer/warning that appears after 4 levels of hierarchy inside their sitemap. Despite the positive feedback, the participants did identify several areas for improvement, including the absence of manual reordering functionality for sitemap cards and the inability to zoom or pan through the canvas when navigating beyond ten nodes.
- The **HCI Persona** tool The HCI Persona tool was the students' favorite due to its live preview and professional output, thanks to our structured fields (demographics, goals, pain points, frustrations). The most requested improvements were improving the quality of image exports, implementing more granular data validation (such as birth date selection), and allowing work and switching between multiple persona types.
- The volunteers appreciated the **HCI Research** tool for its ability to organize and synthesize complex data. The prototype's split between research types proved to be an inspired choice. As volunteers added more data, the tables grew more complex, making the experience more burdensome. A card layout view was mentioned as an alternative to horizontal scroll tables. Moreover, users needed help getting used to the terminology introduced for each user research approach.
- Students reacted positively that they did not actually need to draw using the **HCI Story** prototype. Interacting with the asset collection felt natural and easy to use. The participants identified issues of various UI components – to be addressed in the next iteration of tool development.

5.3. Group 2026: Testing/Evaluating Existing Design Tools

We calculate the SUS values for the existing design tools that the students tested and evaluated (Table 3). Across the three tools evaluated, all mean SUS values fell within the "Good" range (Grade B). Miro Flowcharts scored highest ($M = 84.09$, $SD = 13.82$, 95% CI [80.69, 87.49]), followed by Story Tribe ($M = 79.28$, $SD = 14.82$, 95% CI [75.64, 82.92]) and Slickplan ($M = 76.44$, $SD = 18.10$, 95% CI [71.99, 80.89]), all with $df = 65$.

Compared with the Mean SUS of our AI-generated counterparts (HCI Story, HCI Flow, and HCI Map), their individual SUS values and their mean SUS value (**85.20**) are higher. Across the three HCI AI tools evaluated, all mean

Table 3: SUS values for existing design tools

Tool	StoryTribe	Miro Flowcharts	Slickplan
Mean SUS	79.28	84.09	76.44

SUS values fell within or above the "Good" range. HCI Map achieved the highest score and the only "Excellent" rating (Grade A) ($M = 86.41$, $SD = 10.62$, 95% CI [82.97, 89.85], $df = 38$), followed closely by HCI Flow ($M = 84.93$, $SD = 13.13$, 95% CI [80.62, 89.25], $df = 37$) and HCI Story ($M = 84.24$, $SD = 10.27$, 95% CI [81.08, 87.41], $df = 42$), both rated "Good" (Grade B).

Table 4 lists a comparison of all mean SUS values. *Our AI-generated tools, selected for the usability testing sessions, performed better than the existing tools specific to the student projects' design tasks* (creating storyboards, managing user flows, building sitemaps).

Table 4: SUS comparison between Storyboard, User Flows, and Sitemap (AI-generated & existing design tools)

Story	**Flow**	**Map**	**mean SUS for AI Tool**
84.24 ↑	**84.93** ↑	**86.41** ↑	**85.20** ↑
StoryTribe	Miro Flowcharts	Slickplan	AI-based design tool mean SUS
79.28	84.09	76.44	79.28

Regarding other insights and feedback the students provided for StoryTribe, Miro Flowcharts, and Slickplan, we consider worth mentioning the following:

- Students appreciated StoryTribe because it required no drawing skills; drag-and-drop and the canvas make it accessible to non-designers. Moreover, customization options such as character positions and face traits were a major highlight. Respondents also highly rated StoryTribe's onboarding process. Conversely, volunteers reported usability issues with touch devices, particularly tablets, and encountered visual clutter due to the potential for an overwhelming number of options and features.
- Miro Flowcharts received high marks for its simple, clean, and minimalistic UI and features. Students appreciated not needing an account and the ability to start right from the homepage. Drag-and-drop felt natural and required no learning curve. As disadvantages, respondents noticed confusing UI shifts (with toolbars moving independently of their control) and specific settings or actions breaking their flow.
- Slickplan received the most polarized feedback of the three. While students recognized it as a "professional" and "complete" tool for specific Web architecture tasks, they also criticized it for a steep learning curve and the most restrictive trial model. The snap-to-place feature for sitemaps was a favorite, with students noting it was far more efficient than manually aligning boxes in general design tools. While optimal in some respects, the sheer number of buttons and indicators made the first-time experience intimidating for some.

5.4. Group 2026: Questionnaire

Analyzing the Group 2026's questionnaire results revealed constructive insights. Following the same metrics as Group 2025, the mean Motivation rating is **3.36** ("How does AI affect your motivation to learn?"), while Helpfulness increased significantly to **4.47** ("How helpful was AI in completing your school projects?").

Based on their feedback, the students' increased AI Helpfulness is directly influenced by the now-available wider range of AI tools they can choose from (plus the fact that is no longer just an "extra tool") and by the need to validate/verify the details of generated content. The growing number of LLM models and alternatives leads to more cautious behavior as students verify AI tool outputs more often before accepting them. The questionnaire also provided valuable insights into the current state of AI use among students in the learning and academic context. We highlight below the main observations:

- A large majority (~90%) consider that *a university should include AI training*, how to prompt, how to combat hallucinations and bias, and understanding the ethics of AI. Students believe AI tools are becoming a mandatory industry skill and should prepare accordingly. They do not see AI training as a separate course, but as an integrated module in existing curriculum.
- Students are becoming more cautious about when to use AI tools: if a task is "interesting" or "fun", they do it manually. If it is "uninteresting", "repetitive", or "boilerplate" (e.g., writing a simple CSS class or a standard loop), they delegate it. Moreover, if the goal is to master a subject for their career, they avoid using AI tools to better assimilate new knowledge. If the goal is to "check a box" for a university requirement that they do not value, they use AI to finish it quickly. Students are wary of using AI for high-precision tasks (Math, low-level programming, Machine Learning) because they have learned that AI tools "hallucinate" or give "average/mediocre" answers. Other factors the students mentioned for using AI tools were the tight deadlines of their course assignments and the feeling of an obligation to use it to keep up with the class's collective pace. Pricing is also a crucial factor, as student budgets are often limited, and they exercise caution when making commitments due to the rapid consumption of AI credits.
- ChatGPT, Claude, and Gemini also keep their top ranks for the 2026 group, with slight changes to specific needs: Copilot saw increased usage, while some students mentioned Cursor as an alternative to v0 for direct, high-fidelity prototyping via coding. Another growth spike was observed for DeepSeek, driven by its competitive position with ChatGPT.
- In terms of tasks, the most cited use for AI tools by students is Code Generation (90%), followed by Documentation (70%) and Research (60%). In summary, a student-adopted AI workflow would be as follows: understanding the problem via Research, building the solution with Code Generation, and explaining the work with Documentation and Debriefing. Important to mention that students are *using AI tools at every single stage of the project lifecycle, not just for a quick fix at the end*.
- The biggest advantage of using AI tools is time-saving: time spent explaining requirements, solving repetitive tasks, identifying critical errors, and writing documentation/technical specifications. From a downside perspective, students find they can *produce code that works without actually understanding why*. Students noted that AI tools often provides incorrect or fabricated answers with absolute authority, leading to "debugging hell" where they spend more time fixing AI errors than they would have spent writing the code from scratch. Several students mentioned that *AI tools tend to favor "average" or "standard" solutions*, leading to a loss of creativity and a tendency to accept the first (often mediocre) answer provided.

6. Discussion

Limitations: For our research study, we want to address the following limitations. Firstly, we generated the design tools in the context of the HCI class, its structure and curriculum, and prioritized working with students. Individuals employed did not provide additional insights, as all groups were briefed with the same HCI knowledge during class. Nevertheless, the research still leaves the possibility of expanding its scope and audience. Secondly, we designed and developed independent design tools for students to select based on their needs without additional cognitive load. However, users must synchronize tools and outputs when collaborating on shared projects. Thirdly, the design tools are limited in complexity and are bound to simple, specific use cases (creating user flows, sitemaps, and storyboards). Participants expressed a desire for more advanced features to tackle complex tasks.

Further work: Based on data and feedback, we can extend the study in several directions. We will prioritize feature requests and implement quality-of-life improvements in the tools. Afterward, we intend to consolidate all design tools under a unified Web platform – a project design deliverable smart environment – where students can develop their projects by easily creating the same type of deliverables (research gatherings, ideation solutions, persona profiles, storyboards, user flows, sitemaps, digital sketches, and style guides) required by modern HCI engineering tasks [10].

7. Conclusion

We allowed students full freedom to use AI throughout the course and built design tools for the students' projects, thereby improving engagement, motivation, and course quality. Moreover, when we tested against existing design

software, the AI-generated counterparts demonstrated improved usability – as evidenced by higher SUS values during the research: **85.20** for AI design tools *versus* **79.28** for existing design tools – and usefulness for rapidly producing solutions to specific design challenges. As well, with the rapid adoption of AI in other courses or academic areas, AI-generated tools can upgrade the learning experience [11, 13].

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